GRTE – 4 LAZY F RANCH CABINS

Permit to Construct Application
Engineering Design Report

Prepared by:
Morrison-Maierle



CERTIFICATE OF ENGINEER

State of Wyoming)
) ss
County of Teton)

I, Celine Saucier, do hereby certify that this report was prepared by me, or under my direct supervision.

Bozeman Office
2880 Technology Blvd West
Bozeman, MT 59718

Morrison Maierle
engineers ▪ surveyors ▪ planners ▪ scientists

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GRAND TETON NATIONAL PARK
REHABILITATE 4 LAZY F RANCH CABINS
ENGINEERING DESIGN REPORT

1 SYSTEM INFORMATION

Water System Name:	GTNP – Moose/Beaver Creek
Location:	Moose, Grand Teton National Park, WY 83012
EPA PWS Number:	WY5680093
Population Served:	22
Avg Daily Water Demand:	38 x 125 per capita gpd = 4,750 gpd (WY DEQ minimums)
Max Daily Water Demand:	38 x 340 per capita gpd = 12,920 gpd (WY DEQ minimums)
Source Water Type:	Surface water

2 SCOPE AND PURPOSE

The project will rehabilitate multiple buildings at the former "4 Lazy F" Ranch (4LF) to allow for adaptive reuse of the historic structures as seasonal housing, addressing the park's housing shortage. It will preserve and rehabilitate sleeping cabins, the main lodge, a garage, and outbuildings to provide living quarters, site improvements and utility upgrades for seasonal employees.

The subject property is currently not in use. The existing onsite well and septic facilities will be abandoned and the renovated buildings will be connected to water and sewer mains from Moose, WY. The water and sewer facilities are operated and maintained by the National Parks Service to serve park facilities such as 4LF, and there is sufficient capacity to connect the property with no negative impact to other customers or to treatment and distribution infrastructure.

The property will be occupied during summer months only and will be winterized when not in use. The project will include the installation of a new sanitary sewer collection system, a sewage lift station and force main, and associated equipment and appurtenances between the historic 4LF cabins and the existing sewer treatment facility in Moose, WY. The project also includes the installation of a potable water distribution system for domestic and fire service use connecting to an existing 8-inch water main at the 4LF Site.

In summary the project includes the following site improvements to facilitate the construction of the dormitory:

- Site grading
- Parking lot expansion including ADA spaces
- Sanitary sewer improvements connecting to the existing sewer network
- Water improvements to supply fire protection and domestic water to the cabins

3 BACKGROUND INFORMATION

Project Location

The existing L4F cabins are located at the end of 4 Lazy F Ranch Road north of Moose, WY, approximately one mile north of the Grand Teton National Park Headquarters. The project will rehabilitate 12 buildings on site, 9 of which will provide housing for employees.

The following are the proposed occupancies for the rehabilitated buildings:

- River Cabin – 1 room (2 people)
- Nurse's Cabin – 2 rooms (4 people)
- George's Cabin – 2 rooms (4 people)
- End Cabin – 2 rooms (4 people)
- Middle Cabin – 2 rooms (4 people)
- Mrs. Oliver's Cabin – 2 rooms (4 people)
- New Cabin – 1 room (2 people)
- Harry's Cabin – 1 room (2 people)
- Main Cabin – 2 rooms (12 people)

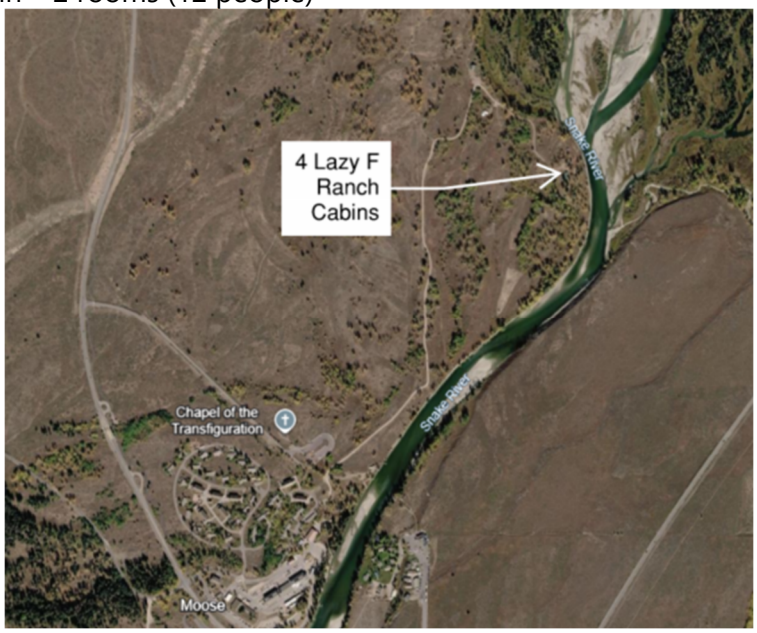


Figure 1. Location of the 4 Lazy F Ranch

Floodplain Considerations

The subject property is shown on FEMA FIRM Panel 2490E, map number 56039C2490E. The 4LF site is within the Special Flood Hazard Area (SFHA) Zone A and is subject to inundation by the 1% Annual Chance Flood. No base flood elevation has been determined by FEMA for the subject property.

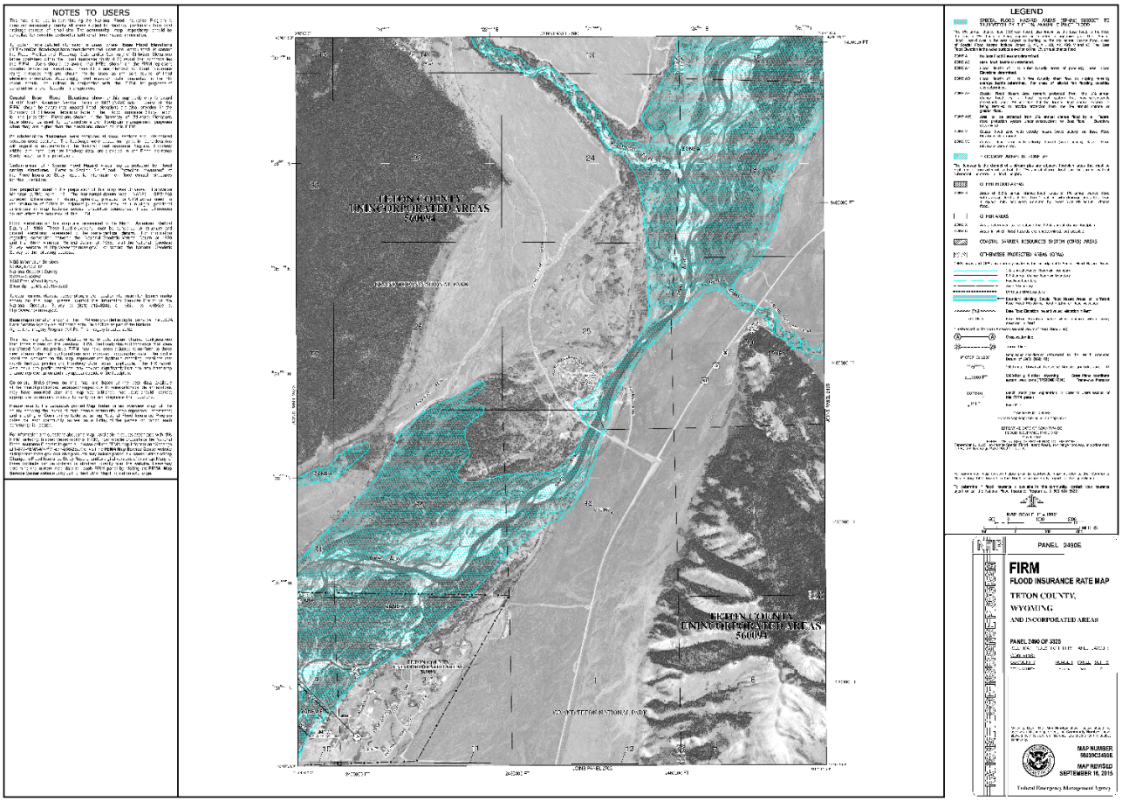


Figure 2. 4 Lazy F Ranch FIRM. Zone A indicates land within the 100-year floodplain with no base flood elevations determined.

Because no base flood elevations have been established for the Zone A Special Flood Hazard Area (SFHA), a site-specific HEC-RAS 2D model was developed to better characterize flood risk at the subject property.

The model terrain was developed using the publicly available digital elevation model (DEM) from the WY_GrandTetoneNP_D22 lidar collection project, published January 29, 2025. As a conservative modeling approach, the water-surface elevations captured during the lidar survey were retained in the model terrain. Based on a comparison of the lidar collection dates with recorded discharge at the USGS stream gauge along the Snake River at Moose, Wyoming (Station 13013650), the retained water surface represents approximately 2,000 to 3,000 cubic feet per second (cfs) of channel flow.

Discharges for the 1-percent annual exceedance probability (AEP) flood, commonly referred to as the 100-year flood, and the 4-percent AEP flood, commonly referred to as the 25-year flood, were obtained from the USGS StreamStats gage information for the Snake River at Moose, Wyoming. The regulated flood-frequency statistics used in the model were:

- 1-percent AEP flood: 24,240 cfs
- 4-percent AEP flood: 20,090 cfs

A spatially varying surface-roughness layer was developed using the 2025 Multi-Resolution Land Characteristics Consortium land-cover dataset at a 30-meter resolution. Manning's roughness coefficients were assigned to the applicable land-cover classifications based on calibrated roughness values reported by Nelson and Schmidt (2007) in Hydrology and Geomorphology of the Snake River in Grand Teton National Park.

The site-specific modeling indicates that the Snake River channel and adjacent floodplain terrace contain modeled flood events up to and including the 1-percent AEP event. The proposed lift-station location is approximately 3.8 feet above the modeled water-surface elevation for the 1-percent AEP event and approximately 300 feet horizontally from the modeled inundation boundary. See Appendix A for an exhibit showing the project location with respect to the 1- and 4-percent AEP events.

These results indicate that the proposed facilities are outside the modeled 1-percent AEP inundation extent and are elevated above the corresponding modeled flood elevation.

4 POTABLE WATER DESIGN

In 2018, an 8-inch HDPE water transmission main was installed from the Moose Water System to 4LF. This main currently dead ends at a fire hydrant near the main lodge. This main has fire suppression capacity to the fire hydrant. Replacement of the internal looped water system will ensure sufficient water and fire service capacity is provided to each of the structures.

The water system will be constructed to meet WYDEQ Water Quality Regulations Chapter 12 requirements to ensure that 500 gpm can be provided by the existing fire hydrant with a 20 psi minimum residual in the distribution system. Each of the water services to the structures are to be constructed to adequately provide fire flows to sprinklers as required in the International Fire Code (IFC) – Site Fire Flow Requirements. Building sprinkler demands are between 30 and 56 gpm per structure. All mains and water services will be constructed at a minimum depth of 6 feet to address freezing concerns with operational blow-offs and pump out drains provided for operations. A new water meter with digital readout will be supplied in a new vault within 4LF area to allow operators to monitor water use.

The new water distribution system shall be looped, with service lines to each plumbed structure stemming off the water main. Each service line shall include an isolation valve and shall be sloped back to the water main or to the structure to allow for draining and winterization. Isolation valves shall be included at each branch of the water main and at the midpoint of the looped main.

New main and service lines shall be constructed of HDPE and equipped with tracer wire to allow location. New water mains shall be 4" DR-11 HDPE meeting AWWA C906 standards. All new service lines constructed shall be sized appropriately for any applicable fire suppression flows. New service lines shall not exceed 200 feet. Air relief valves will be furnished at all high points in the system with drains at all low points.

The existing water well on site will be abandoned per WYDEQ Water Quality Regulations Chapter 26. Following abandonment, a report will be required by the State of Wyoming Engineers Office, Ground Water Division. Due to the historic nature of 4LF, the existing well house fabric will be documented in place, set aside for future reconstruction. All other existing well system infrastructure will remain abandoned in place.

Average and maximum day potable water demands were calculated using DEQ Chapter 12, Section 10(i) values while peak hour rates use a multiplier of 3.4 applied to the average flows, as presented in Table 1 . These values are considered very conservative as there is no irrigation potential off this supply line.

Table 1. Projected potable water demands.

Average Day	**Max Day**	**Peak Hour**
125 GPDPC*38 people = 4,750 GPD=3.3 GPM	340 GPDPC*38 people = 12,920 =8.97 GPM	3.3 GPM * 3.4 =11.22 GPM

5 WASTEWATER DESIGN

A new collection system will be constructed to serve the cabins. A lift station constructed in a central location in the 4LF area will pump this wastewater through a force main to the existing wastewater collection system and treatment plant in Moose.

Estimated flows were developed using the same per-capita values as water demands, resulting in the flows listed in Table 2. A peaking factor of 2.8 was used for maximum day flows. A peaking factor of 4.34 was used for peak hour flows based on population served.

Table 2. Projected wastewater demands.

Average Day	**Max Day**	**Peak Hour**
125 GPDPC*38 people = 4,750 GPD=3.3 GPM	3.3 GPM * 2.8 = 9.24 GPM = 13,306 GPD	3.3 GPM * 4.34 = 14.3 GPM = 20,624 GPD

6 GEOTECHNICAL REPORT

A geotechnical report was not provided as a part of this project.

7 WYDEQ CHAPTER 11: DESIGN AND CONSTRUCTION STANDARDS FOR SEWERAGE SYSTEMS

For this project, Chapter 11 of the *Wyoming DEQ Design and Construction Standards* was reviewed:

Effective Date of Chapter 12: 06/29/2018-Current
Reference Number: 020.0011.11.06292018

Section 6. Engineering Design Report

This report is provided in compliance with the guidelines of this section.

Section 7. Plans and Specifications Content.

Plans and specifications have been provided meeting the guidelines of this section.

Section 9. Design of Sewers

9.a No combined or underground stormwater facilities exist or are proposed. All stormwater is controlled through surface features.

9.b 8-inch SDR35 PVC is specified for use as sanitary sewer for gravity collection lines and 4-inch HDPE is specified for use for the force main. Sewer services will be 2" HDPE. Pipe materials and pressure class ratings are selected to resist expected soil, groundwater, and wastewater conditions and loading rates. All joints will be flexible, durable, and watertight. Leakage tests will be conducted prior to certification.

9.c Collection piping

- ***i.*** Gravity system. Sewer depths will be between four and six feet. The system will be drained and winterized and will not be subject to freezing conditions when in use. The 8-inch sewer will be installed with a minimum slope of 0.5% with a straight alignment between manholes. Minimum velocity at full flow is 3.5 feet per second. Excavation, bedding, and backfill specifications comply with the applicable requirements of this section.
- ***ii.*** Force mains and pressure sewers. Minimum installation depth without board insulation is six feet. The force main will be 4-inch HDPE. Minimum flow velocity at the design operating point is 2.55 feet per second. Air relief valves will be located at all high points in the force main. Cleanouts will be installed at a maximum spacing of 400 feet. Pressure sewer service lines will be 2" in diameter and preceded by grinder pump stations. Pumps will include isolation and check valves and tanks will be non-metallic polymolded material with a minimum chamber volume of 50 gallons.
- ***iii.*** Gravity services will be 4" in diameter, installed at a minimum 2% slope, and connected to the gravity main via wye or tee connection.

9.d Manholes shall be located at all changes in alignment and at intersections and ends, with spacing not to exceed 400 feet. Manholes shall have a minimum inner diameter of four feet. Manholes will be precast concrete. Covers will be watertight with a minimum opening of 22-inches. Design of manholes will conform with all guidelines of this section.

9.e Special structures – not applicable.

9.f Potable supply protection – there will be no cross-connections. All requirements for separation will be met where the proposed sewer crosses water mains. A cement-treated slurry will be used if a separation of at least 18-inches cannot be maintained.

Section 10. Pumping Stations

10.a Design conditions

- ***i.*** Total dynamic head of the pumping units at the design operating flow of 82 gpm per pipe is 74 feet, based on pipe friction, losses, and static head.
- ***ii.*** Grit removal is not provided. Pumps and pipe assemblies are suitable for raw wastewater handling.
- ***iii.*** No screening is required. Average day flow is 4,752 gpd.
- ***iv.*** Pumps will be Flygt NP 3085SH 3 Adaptive 255. Pumps can handle 3" solids. Discharge piping and appurtenances shall be 4".
- ***v.*** The lift station is sized so that pump cycle time is 35 minutes at average day demand, and 9 minutes at maximum (peak hour) flow conditions.
- ***vi.*** All submersible wet well equipment will be installed on guide rails for removal without wet well entrance.
- ***vii.*** Pipes and appurtenances will be reinforced as necessary to resist the maximum possible surge control.
- ***viii.*** NPSHa at maximum operating flow is 25.7 feet. NPSHr for the selected pump is 17.6 feet.
- ***ix.*** The 5-foot diameter wet well will be constructed of reinforced fiberglass with a poured-in-place concrete foundation to serve as a ballast resisting hydrostatic uplift.

10.b Siting requirements

- ***i.*** The lift station wet well, valve enclosure, and controls will be accessible to operators at all times.
- ***ii.*** Flood protection. The 4LF property is within the delineated 100-year floodplain. The area surrounding the lift station will be graded so that the wet well hatch, valving, and controls will not be subject to standing water or flowing surface runoff. A site-specific 2D HEC-RAS model was developed which determined that the existing ground elevation of the lift station site is 3.8 feet above the 100-year base flood elevation.

10.c Pump station Types

- ***ii.*** The wet well is designed to prevent vortexing or unstable pumping conditions. Pump intake is below the minimum water level. Hopper walls will be sloped as required. Provisions for bypass, isolation, and/or dewatering will be provided.
- ***iii.*** Submersible pumps are removable without wet well entrance and will be provided with motor seal failure indicators. All electrical equipment within the wet well will be rated for Class 1 Div 1 installation.
- ***vi.*** Two packaged grinder pump assemblies will be installed at individual cabins to convey wastewater where minimum service line slope cannot be met. These grinder pumps will serve only one residential cabin each.

10.d Piping and valves

i. Pump intake will be 0.5-feet below the minimum water level and located on the opposite wall of the wet well from the gravity sewer inlet.

ii. Force main piping will be 4" with a minimum velocity of 2.55 feet per second. The force main will be blocked and restrained against surging and thrust movement as required.

iii. Valves will be located in an above-grade enclosure adjacent to the wet well. The enclosure will be sited to protect against flooding. Pump discharges will have shut-off valves and check valves. Air relief valve vaults will be installed at the high point along the force main.

10.e Reliability

i. Two pumps will be provided. Design operating conditions at maximum design flow are attainable with a single pump in service.

ii. A portable generator or pump is available through the wastewater treatment and collections operating district for emergency power. The wet well is sufficiently deep such that emergency response time at average flow is 30 hours.

10.f Electrical

i. All non-submersible motors and controls will be installed in an enclosure or panel that will be sited to remain undamaged by the 100-year flood.

ii. Each pump will include manual and automatic start and stop controls.

iii. Electrical design and specifications will comply with NEC as adopted and amended by the Wyoming Department of Fire Prevention and Electrical Safety. Electrical components located in the wet well will be rated for Class 1, Div 1, Grade C/D installation.

iv. High- and low-level alarms within the wet well will be linked to a panel-mounted RTU and incorporated into existing SCADA overseen by the Moose treatment plant operators.

10.g Safety

i. The wet well is not designed for normal access. A portable blower or gravity ventilation can be utilized if necessary for safe entrance.

ii. Hoists for equipment removal shall be provided by the receiving treatment plant operations.

iii. Portable lighting shall be provided by maintenance and operations staff if access is required outside of daylight hours.

iv. No moving equipment requiring shields or guards will be accessible.

v. Warning signs for electrical hazards and other unsafe conditions will be provided.

vi. The lift station will comply with applicable Wyoming Occupational Health and Safety Rules and Regulations.

Sections 11-22

Wastewater from the site will be sent to the facility at Moose WY which is located approximately one mile south of the site. No improvements or alterations are being made to the treatment facility. These sections are not directly related to the work included in this project.

Section 23. Laboratory Requirements

Testing required for the project will be conducted through a certified laboratory.

Section 24. Operation and Maintenance Manuals

Any operation and maintenance manuals for equipment installed as part of the project will be provided to the operators of the GTNP – Moose/Beaver Creek public water system.

8 WYDEQ CHAPTER 12 DEQ RULES & REGULATION FOR PUBLIC WATER SUPPLIES

For this project, Chapter 12 of the WY DEQ Design and Construction Standards was reviewed. Below is a summary of applicable items.

Effective Date of Chapter 12:	07/26/2023-Current
Reference Number:	020.0011.12.07262023

Section 7. Permits, Application, and Recordkeeping

Design reports, plans, and other pertinent information have been provided meeting the guidelines of this section.

Section 8. Plans and Specifications

Plans have been provided meeting the guidelines of this section.

Section 9. Engineering Design Report

This report is provided in compliance with the guidelines of this section.

Section 16. Distribution Systems

16.a System design complies with all applicable and required provisions of the 2018 Ten States Standards as listed in WYDEQ Chapter 12, Section 16.

16.b Pipe requirements. 4-inch C906 SDR-11 HDPE is specified for this project conforming with the required guidelines.

16.c Flanged fittings will not be used for the water distribution system.

16.d Hydraulic analysis. Hydrant test data collected by the NPS on July 14, 2026 was utilized to develop a hydraulic model of the proposed system. See Chapter 9 of this report for a discussion of the system water model and hydraulic results.

- ***i.*** As modeled, the minimum pressure in the system at maximum day demand plus fire flow is 49 psi
- ***ii.*** The normal working pressure of the system as modelled is 73-78 psi.

16.e The 4-inch water main has been evaluated for capacity to serve fire flow as determined by the fire protection engineer.

16.f No new hydrants are specified.

16.g Not applicable, the largest new water main will be 4".

16.h Excavation and trenching will be conducted as required by this section.

- ***i.*** The trench bottom elevation shall be excavated for the bell of the pipe
- ***ii.*** Rock shall be removed within 6" of the pipe
- ***iii.*** Trenching shall be dewatered for all work as required

16.i Pipe bedding shall conform to ASTM D2321 Class 1, 2, or 3.

16.j Pipe fittings shall be watertight and comply with AWWA M55 requirements for the installation of HDPE pipe and fittings.

16.k Backfill will be performed without disturbing the pipe alignment and will use only suitable material compacted to a density greater than that of the surrounding soil.

16.l Separation of water mains from sanitary and storm sewers. The new water main will be constructed over 10 feet from all sanitary sewer mains. No storm sewers are to be constructed.

16.m Cross connections. The fire suppression lines inside of buildings shall be equipped with an RPZ backflow preventor device to prevent any cross-connection from the fire suppression system. Backflow prevention is shown on sheet 000 F004 as an 1.25" Ames 2000B-FP double check valve assembly. No cross-connections will exist between the potable water system and any other pipes or conveyances.

Section 17. Laboratory Requirements

Testing required for the monitoring and analysis required for this project will be conducted through a certified laboratory.

Section 18. Operation and Maintenance Manuals

Any operation and maintenance manuals for equipment installed as part of the project will be provided to the operators of the GRTE-Moose public water system.

9 HYDRAULIC ANALYSIS

National Parks Service engineers conducted a two-hydrant flow test on July 14, 2026 at approximately 2:20 pm, resulting in the following:

- Test Hydrant: SE (at 4LF)
  - Location: 43°40'0.71"N, 110°42'20.01"W
- Residual Hydrant: NW (at Caretakers Cabin)
  - Location: 43°40'8.07"N, 110°42'30.41"W
- Outlet Diameter: 2.5"
- Static Pressure: 73 psi
- Residual Pressure 29 psi
- Calculated Flow: 705 gpm

This data was used to model available flow to the development as a reservoir-and-pump proxy. A pump curve was established utilizing the emitter discharge equation and NFPA 291 Equation 4.10.1.2. The system was modeled in Bentley WaterGEMS 23.00.00.19.

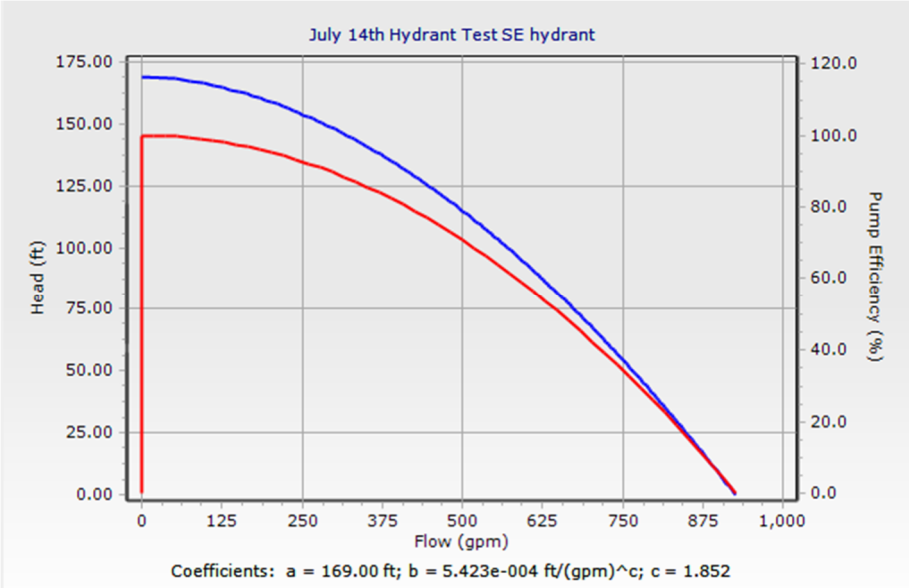
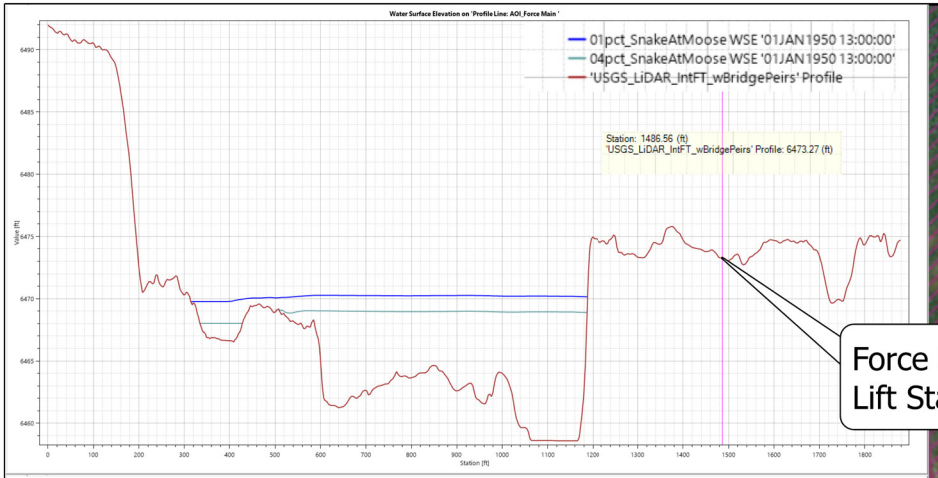


Figure 3 Hydrant pump curve for hydraulic model

Scenarios were calculated for average-day demand and maximum-day demand plus fire flow. See Appendix C for tabulated model results. Static pressure in the system is expected to be 73-78 psi. Iterative fire flow analysis found that all building fire suppression systems can provide the required flow at maximum-day demands and that the existing hydrant can provide up to 770 gpm without system pressures dropping below 20 psi. Maximum pipe velocity at fire flow is 8.82 feet per second.

APPENDIX A – BASE FLOOD ELEVATIONS AND FLOODPLAIN PROFILE FROM 2D HEC-RAS MODELING





Force Main Lift Station

Force Service Lift Station

Zone A

HEC-RAS 2D
01 pct Inundation

Snake River

Blacktail Ponds Overlook

Zone A

Zone A

Snake River

4 Lazy F Ranch Rd

Legend

- Lift Station
- Proposed Sanitary Sewer System
- Profile Line
- InundationBoundary_01pct
- InundationBoundary_04pct
- Mapped Zone A

0 500 1,000 2,000 Feet

N

Morrison Maierle
engineers · surveyors · planners · scientists

2880 Technology Blvd. W
Bozeman, MT 59718

(406) 587-0721

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HEC-RAS 2D INUNDATION MAP

GRTE - LAZY F RANCH CABINS

PROJECT NO.
620401.002

APPENDIX A

APPENDIX B – LIFT STATION DESIGN AND HYDRAULIC CALCULATIONS

GRTE LAZY F RANCH - LIFT STATION
SEWER FLOW CALCULATIONS

Project #: 630401002
Date: 6/15/2026
Preparer: APR

SEWER FLOW CALCULATIONS

Assumptions and Background:

- Location: Moose WY
- 12 total buildings (9 for staff housing)
- Residential buildings occupancy = 38 residents (3x2, 5x4, 1x12)
- Is within a flood zone (A)
- Moose WY WWTP will be pumped to it via HDPE FM
- ADD 3.3 gpm
- MDD 9.24 gpm
- PHD 92 gpm
- Chapter 11 WY DEQ
- Main Lodge Total Peak Flow by Fixture Count 19 gpm
- Main Lodge sump: 20 gpm/15 ft head - may not be connected to sanitary sewer
- 4-6 foot burial depth (exst) - not used in winter. 8" 0.5% min slope
- Force main alignment was previously evaluated and selected
- two smaller packaged grinder lift stations - at Caretakers and southernmost cabin ("New Cabin")
- limited fiber optic SCADA to Moose WWTP
- Min SCADA communications: wet well level, pump runs/faults, HA and LA
- Concrete wet well and above-ground valve building/enclosure
- discharge header valves above ground
- press-seal pipe penetration gaskets
- hinged metal access hatch 36"x60" min
- 20 mil epoxy coating
- two submersible pumps, guide rails, vent, etc
- capacity: accept peak hour flows, max pump cycling 1 every 10 min
- storage in wet well: so that 30% above max flow can be accepted without exceeding 6 cycles/hr
- 3" min solids passing
- appx 6,200 LF of 4" HDPE force main to exst Moose system

LAZY F RANCH LIFT STATION SEWER FLOW CALCULATIONS

(continued)

Total Sewer Flows to Lift Station

Average Day Flow (ADF)	4,752	gpd
Average Day Flow (ADF)	3.3	gpm
Equivalent Population	38	people
Max Day Flow (MDF)	13306	gpd
	9.24	gpm
Peak Factor (PF)	4.34	*(MDEQ 10-1)*
Peak Hour Flow (PF)	14.3	gpm
	20624	gpd

GRTE LAZY F RANCH - LIFT STATION
SEWER FLOW CALCULATIONS

Project #: 630401002
Date: 6/15/2026
Preparer: APR

LIFT STATION PARAMETERS AND CALCULATIONS

Assumptions:
- Elevations and dimensions are derived from POR drawing set
- Set points listed are designed to meet DEQ requirements and pump manufacturer requirements.

Parameters:

Average Day Flow (ADF)	3.3	gpm
Maximum Assumed Flows	14.3	gpm
Wet Well Diameter:	5.0	ft
Ground Elevation at Lift Station:	6473.5	ft
Rim Elevation at Lift Station:	6474.0	ft
Sewer Main Invert at Lift Station:	6,465.66	ft
Pump Inlet Elevation:	6,460.91	ft
Base of Wetwell	6,460.41	ft
Low Alarm:	6,461.41	ft
Pump Off:	6,461.91	ft
Lead Pump On:	6,462.66	ft
Lag Pump On:	6,463.16	ft
High Alarm:	6,463.66	ft
Gallons Per Foot Depth:	146.9	gallons/ft
Effective Volume Fill time:	33.4	min
Wet Well Storage Volume:	550.8	cu. ft.
Wet Well Storage Volume:	4,120.26	gallons
Emergency Response Time at Ave Flow:	1249	min
Effective Volume (EV):	110.2	gallons
Effective volume depth (5-ft dia.):	0.75	ft

Calculations:

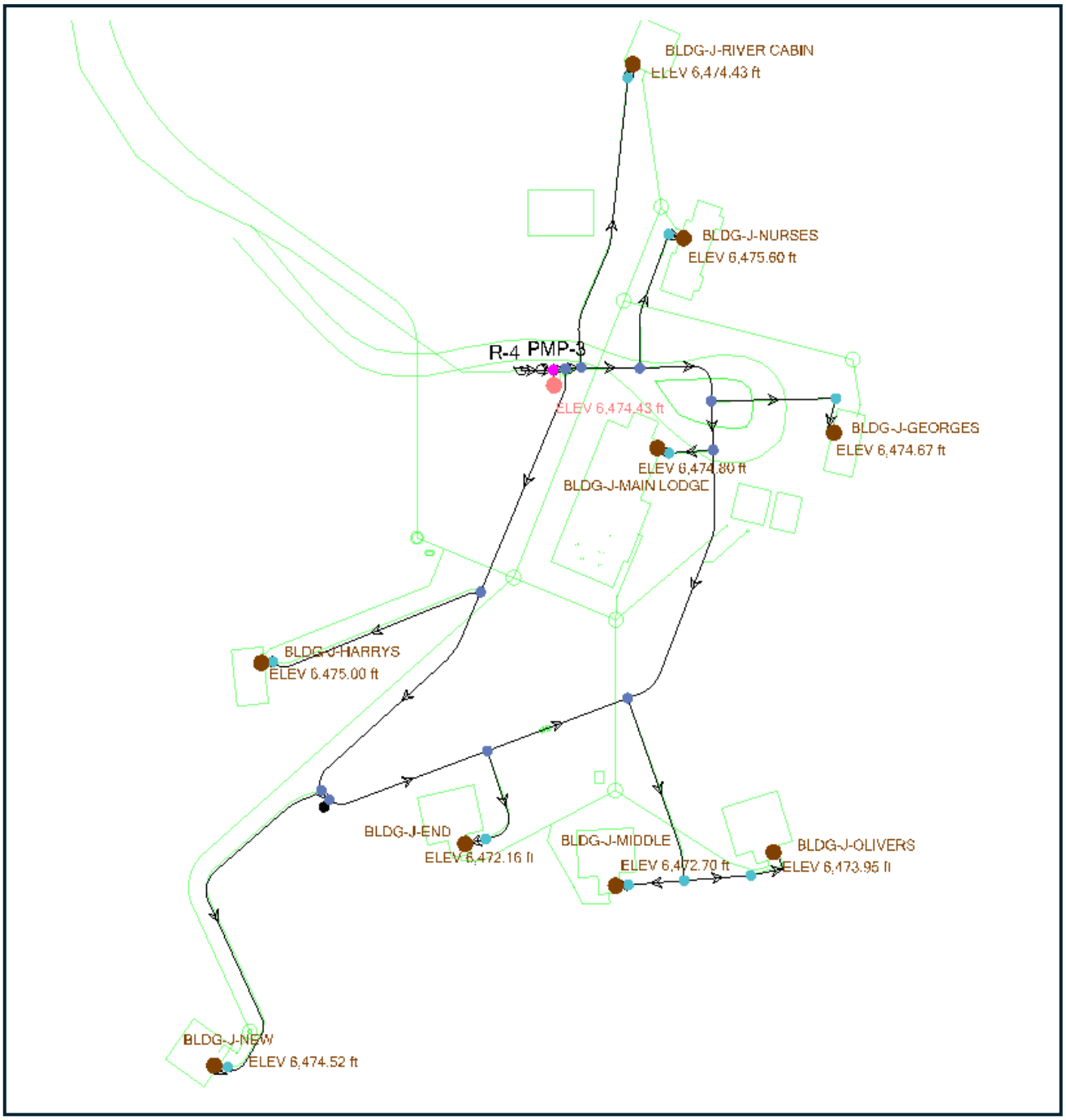
Fill time = EV / ADF	33.4	min
Drawdown rate = Pumping rate - Filling rate	79.4	gpm
Drawdown time = EV / drawdown rate	1.4	min
Pump cycle time = Fill time + Drawdown time	*34.8*	*min*

Therefore, every 34.8 minutes the pumps will cycle during average day flow conditions.
This equates to 2 pump cycles per hour.

Max Flow Fill time = EV / PHF	7.7	min
Drawdown rate = Pumping rate - Filling rate	68.4	gpm
Drawdown time = EV / drawdown rate	1.6	min
Pump cycle time = Fill time + Drawdown time	*9.3*	*min*

Therefore, every 9.3 minutes the pumps will cycle at Peak Hour flow conditions.
This equates to 6.5 pump cycles per hour.

APPENDIX C – WATER SYSTEM MODEL RESULTS AND FIGURES


BLDG-J-RIVER CABIN
ELEV 6,474.43 ft
BLDG-J-NURSES
ELEV 6,475.60 ft
R-4 PMP-3
ELEV 6,474.43 ft
BLDG-J-GEORGES
ELEV 6,474.67 ft
ELEV 6,474.80 ft
BLDG-J-MAIN LODGE
BLDG-J-HARRYS
ELEV 6,475.00 ft
BLDG-J-END
ELEV 6,472.16 ft
BLDG-J-MIDDLE
ELEV 6,472.70 ft
BLDG-J-OLIVERS
ELEV 6,473.95 ft
BLDG-J-NEW
ELEV 6,474.52 ft

Label	Elevation(ft)	Pressure(psi)	Demand(gpm)
BLDG-J-MAIN LODGE	6,474.80	74	1.04
BLDG-J-MIDDLE	6,472.70	75	0.35
BLDG-J-NURSES	6,475.60	73	0.35
BLDG-J-END	6,472.16	75	0.35
BLDG-J-OLIVERS	6,473.95	74	0.35
BLDG-J-GEORGES	6,474.67	74	0.35
BLDG-J-RIVER CABIN	6,474.43	74	0.17
BLDG-J-HARRYS	6,475.00	74	0.17
BLDG-J-NEW	6,474.52	74	0.17
Hyd-J-8	6,474.43	74	0
NewMain-J-10	6,467.61	77	0
XC-J-11	6,467.63	77	0
J-12	6,465.37	78	0
NewMain-J-13	6,465.44	78	0
SL-J-14	6,468.45	76	0
SL-J-16	6,467.00	77	0
SL-J-18	6,464.59	78	0
SL-J-20	6,467.29	77	0
SL-J-22	6,467.02	77	0
SL-J-24	6,468.18	77	0
SL-J-27	6,463.99	78	0
SL-J-29	6,465.78	78	0
SL-J-31	6,467.01	77	0
NewMain-J-32	6,466.70	77	0
NewMain-J-33	6,465.13	78	0
NewMain-J-34	6,466.77	77	0
NewMain-J-35	6,467.72	77	0
SL-J-36	6,465.19	78	0
NewMain-J-37	6,466.62	77	0
NewMain-J-38	6,466.34	77	0
NewMain-J-39	6,465.95	77	0
NewMain-J-40	6,467.64	77	0

Label	Satisfies Fire Flow Constraints?	Demand (gpm)	Fire Flow Needed (gpm)	Total Flow Needed (gpm)	Total Flow Available (gpm)	Calculated Residual Pressure @ Total Flow Needed (psi)	Pressure (Calculated System Lower Limit) (psi)
Hyd-J-8	TRUE	0	500	500	770	50	20
BLDG-J-NEW	TRUE	0.46	56	56	100	73	72
BLDG-J-RIVER CABIN	TRUE	0.46	52	52	100	73	72
BLDG-J-MAIN LODGE	TRUE	3	47	50	103	73	72
BLDG-J-HARRYS	TRUE	0.46	30.4	30.86	100	73	72
BLDG-J-MIDDLE	TRUE	0.95	30.4	31.35	101	74	72
BLDG-J-END	TRUE	0.95	30.4	31.35	101	75	72
BLDG-J-OLIVERS	TRUE	0.95	30.4	31.35	101	74	72
BLDG-J-GEORGES	TRUE	0.95	30.4	31.35	101	73	72
BLDG-J-NURSES	TRUE	0.95	30.2	31.15	101	73	72
NewMain-J-10	TRUE	0	0	0	100	77	72
XC-J-11	TRUE	0	0	0	100	77	72
J-12	TRUE	0	0	0	100	77	72
NewMain-J-13	TRUE	0	0	0	100	77	72
SL-J-14	TRUE	0	0	0	100	76	72
SL-J-16	TRUE	0	0	0	100	77	72
SL-J-18	TRUE	0	0	0	100	78	72
SL-J-20	TRUE	0	0	0	100	77	72
SL-J-22	TRUE	0	0	0	100	77	71
SL-J-24	TRUE	0	0	0	100	76	72
SL-J-27	TRUE	0	0	0	100	78	72
SL-J-29	TRUE	0	0	0	100	77	72
SL-J-31	TRUE	0	0	0	100	77	72
NewMain-J-32	TRUE	0	0	0	100	77	72
NewMain-J-33	TRUE	0	0	0	100	78	72
NewMain-J-34	TRUE	0	0	0	100	77	72
NewMain-J-35	TRUE	0	0	0	100	76	72
SL-J-36	TRUE	0	0	0	100	78	72
NewMain-J-37	TRUE	0	0	0	100	77	72
NewMain-J-38	TRUE	0	0	0	100	77	72
NewMain-J-39	TRUE	0	0	0	100	77	72
NewMain-J-40	TRUE	0	0	0	100	77	72

Label	Length (ft)	Start Node	Stop Node	Diameter (in)	Material	Hazen-Williams C	Flow * (gpm)	Velocity * (ft/s)
HydLead-P-1	8	Hyd-J-8	XC-J-11	8	Ductile Iron	130	0.00	0
P-6	11	R-4	PMP-3	8	Ductile Iron	130	8.98	0.06
P-7	6	PMP-3	XC-J-11	8	HDPE	140	8.98	0.06
P-17NewMain-	131	NewMain-J-10	NewMain-J-38	4	HDPE	140	3.00	0.08
P-18NewMain-	143	NewMain-J-38	NewMain-J-39	4	HDPE	140	2.54	0.06
P-19	7	NewMain-J-39	NewMain-J-13	4	HDPE	140	2.07	0.05
P-20NewMain-	93	NewMain-J-13	NewMain-J-33	4	HDPE	140	2.07	0.05
P-21NewMain-	83	NewMain-J-33	NewMain-J-37	4	HDPE	140	1.12	0.03
P-22NewMain-	150	NewMain-J-37	NewMain-J-32	4	HDPE	140	0.78	0.02
P-23NewMain-	27	NewMain-J-32	NewMain-J-34	4	HDPE	140	3.61	0.09
P-24NewMain-(1)	54	NewMain-J-34	NewMain-J-35	4	HDPE	140	4.56	0.12
P-24NewMain-(2)	32	NewMain-J-35	NewMain-J-40	4	HDPE	140	5.52	0.14
P-25NewMain-(1)	9	NewMain-J-40	NewMain-J-10	4	HDPE	140	5.98	0.15
P-25NewMain-(2)	7	NewMain-J-10	XC-J-11	4	HDPE	140	8.98	0.23
P-26	5	J-12	NewMain-J-13	4	HDPE	140	0.00	0
SL-P-27	24	SL-J-20	NewMain-J-32	4	HDPE	140	2.83	0.07
SL-P-28	69	SL-J-31	NewMain-J-34	4	HDPE	140	0.95	0.02
SL-P-29	161	NewMain-J-40	SL-J-14	4	HDPE	140	0.46	0.01
SL-P-30	123	SL-J-16	NewMain-J-38	4	HDPE	140	0.46	0.01
SL-P-31	75	NewMain-J-35	SL-J-24	4	HDPE	140	0.95	0.02
SL-P-32	57	SL-J-27	NewMain-J-33	4	HDPE	140	0.95	0.02
SL-P-33	202	SL-J-22	NewMain-J-39	4	HDPE	140	0.46	0.01
SL-P-34	31	SL-J-18	SL-J-36	4	HDPE	140	0.95	0.02
SL-P-35	37	SL-J-36	SL-J-29	4	HDPE	140	0.95	0.02
SL-P-36	104	SL-J-36	NewMain-J-37	4	HDPE	140	1.90	0.05
SL-P-37	7	SL-J-16	BLDG-J-HARRYS	4	HDPE	140	0.46	0.01
SL-P-38	7	SL-J-20	BLDG-J-MAIN LODGE	4	HDPE	140	2.83	0.07
SL-P-39	8	SL-J-14	BLDG-J-RIVER CABIN	4	HDPE	140	0.46	0.01
SL-P-40	9	SL-J-24	BLDG-J-NURSES	4	HDPE	140	0.95	0.02
SL-P-41	25	BLDG-J-GEORGES	SL-J-31	4	HDPE	140	0.95	0.02
SL-P-42	29	BLDG-J-OLIVERS	SL-J-29	4	HDPE	140	0.95	0.02
SL-P-43	7	SL-J-18	BLDG-J-MIDDLE	4	HDPE	140	0.95	0.02
SL-P-44	12	BLDG-J-END	SL-J-27	4	HDPE	140	0.95	0.02
SL-P-45	10	SL-J-22	BLDG-J-NEW	4	HDPE	140	0.46	0.01

* Transient Results from Automated Fire Flow not included in Pipe Results Table.

Estimated velocity through at 6in pipe *hydrant lead) at 770 gpm is 8.74 fps

Estimated velocity through a 4in pipe (service line) at 53.4 gpm (maximum MDD+sprinker building demand) is 1.44 fps

Scenario	Elevation (ft)	Pump Definition	Status (Initial)	Hydraulic Grade (Suction) (ft)	Hydraulic Grade (Discharge) (ft)	Flow (Total) (gpm)	Pump Head (ft)
BASE (ADD)	6472	JULY 14TH HYDRANT TEST SE HYDRANT	On	6476	6645	3.3	169
MDD + FF *	6472	JULY 14TH HYDRANT TEST SE HYDRANT	On	6475.93	6577.16	565	101.23

* MDD+FF scenario with hydrant and largest sprinkler system as fixed demands

	Flow (gpm)	Head (ft)
Shutoff:	0.00	169.00
Design:	705.00	67.00
Max. Operating:	780.00	46.00

